SARS-COV-2 NUCLEOCAPSID PROTEIN ENGAGES WITH VIRAL RNA AND ERGIC LIPIDS TO DRIVE VIRAL CORE ASSEMBLY

Shovon Swarnakar[1,2 †], Jhili Mishra[1†], Virgile Rat[1], Peggy Merida[1], Louis Bunel[1], Damien Avinens[1], Jérôme Feuillard[1], Sébastien Lyonnais[3], Eric Martinez[1], Jitendriya Swain[1,3], Jérémy Dufourt[1], Mickael Blaise[1], **Cyril Favard**[1§], Jaydeep K Basu[2§] and Delphine Muriaux[1,3§ *]

1. *Institut de Recherche en Infectiologie de Montpellier, IRIM, CNRS & University of Montpellier, UMR9004, 1919 route de Mende, Montpellier, France*
2. *Department of Physics, Indian Institute of Science, Bangalore, C.V. Raman Road, Karnataka 560012, India*
3. *CEMIPAI, CNRS & University of Montpellier, UAR3725, 1919 route de Mende, Montpellier, France*

Severe acute respiratory syndrome coronavirus 2 assembles at the ER–Golgi intermediate compartment (ERGIC), yet the molecular basis of nucleocapsid (N) protein interactions with host membranes remains unclear. Using in vitro reconstituted lipid membranes and viral RNA –N complexes, we confirm that full-length N binds phosphatidylinositol (PI)- and phosphatidylserine (PS)-containing membranes[1] and, in addition, show that N induces lipid clustering upon self-interacting on membranes. The isolated N-terminal domain of N protein lacks this activity, whereas its C-terminal domain retains membrane-associated self-assembly and lipid sorting. Even on simple lipid bilayers composition, we observe co-clustering of N and PI lipids, facilitating ribonucleoprotein (RNP) assembly on membranes. Furthermore, we show that ERGIC-mimicking membranes enhances PS-RNP co-clustering to dimensions matching the viral core size. Finally, we observed in cells that the presence of viral RNA favours N clustering that can be located on ERGIC. These findings reveal cooperative interactions between N, viral RNA, and ERGIC lipids as key drivers of lipid-dependent viral core formation, providing a mechanistic framework for the early steps of viral assembly.

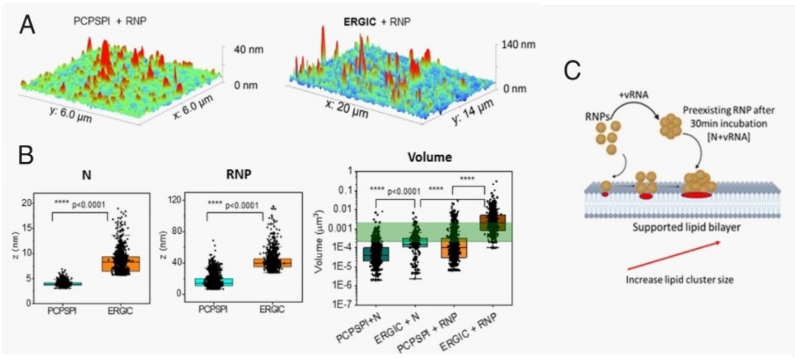

Figure 1 : *A-AFM topology of SARS-CoV2 N RNPs on PCPSPI or ERGIC mimicking membranes B- Height (left) and volume distribution of RNP clusters C- A schematic observation of the membrane RNP formation.*

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2. *Savastano, A., Ibáñez de Opakua, A., Rankovic, M. et al. Nucleocapsid protein of SARS-CoV-2 phase separates into RNA-rich polymerase-containing condensates. Nat Commun* ***11****, 6041 (2020). DOI:10.1038/s41467-020-19843-1*